
The Waiting Game:
A Survival Analysis of Fernando Tatis Jr.'s Most Baffling Power Drought
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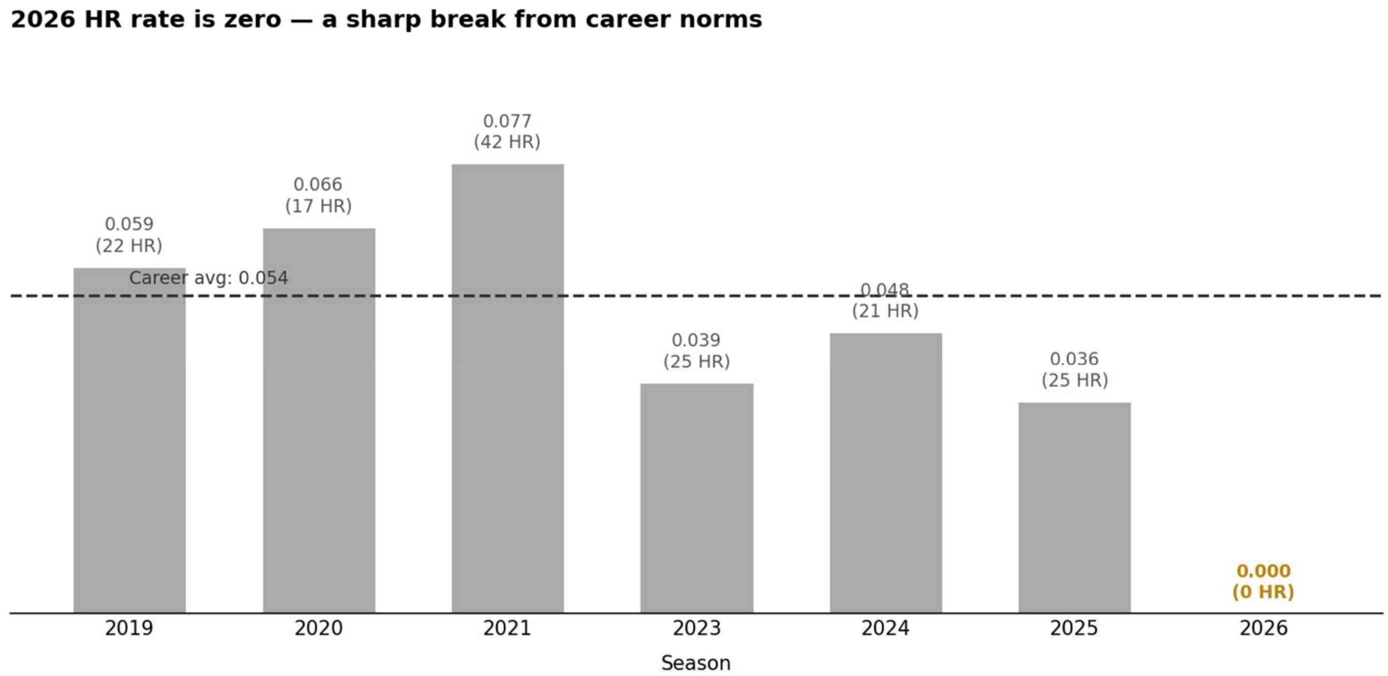
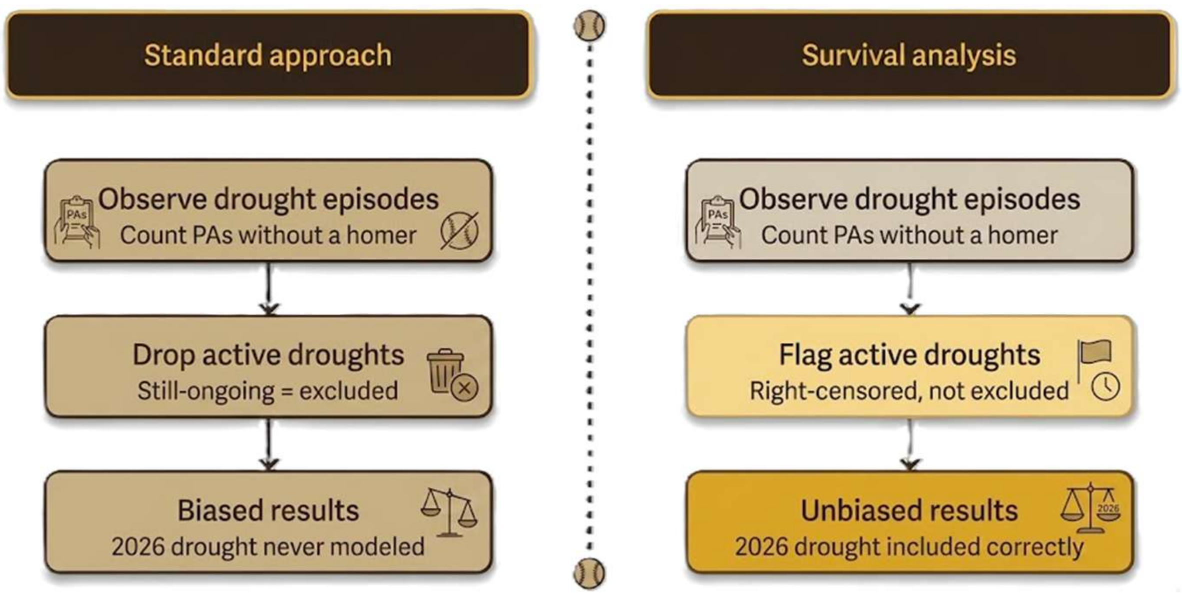
The Problem

- 180+ plate appearances in 2026 with zero home runs
- Exit velocity: 91.6 mph, still elite
- Standard metrics offer no signal: HR count is zero until it isn’t
- Question: Is this bad luck, or is something structurally wrong?



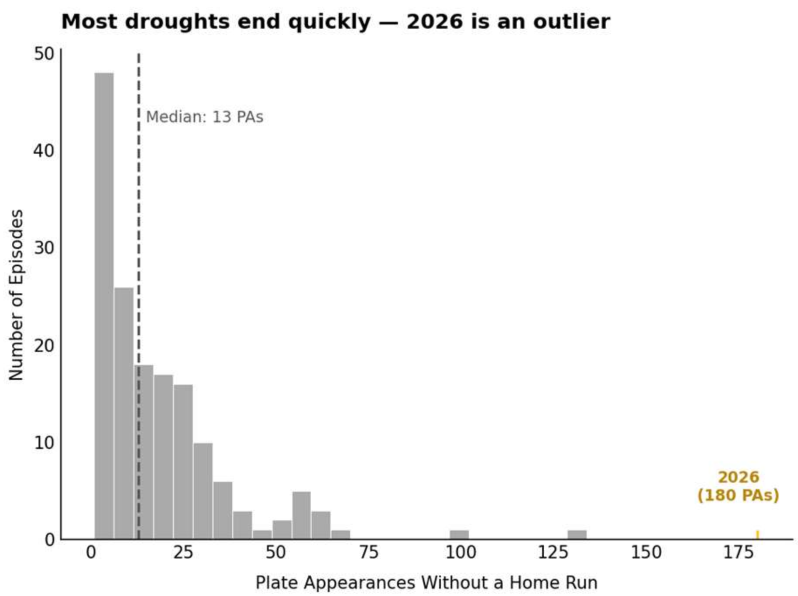
Why Survival Analysis?

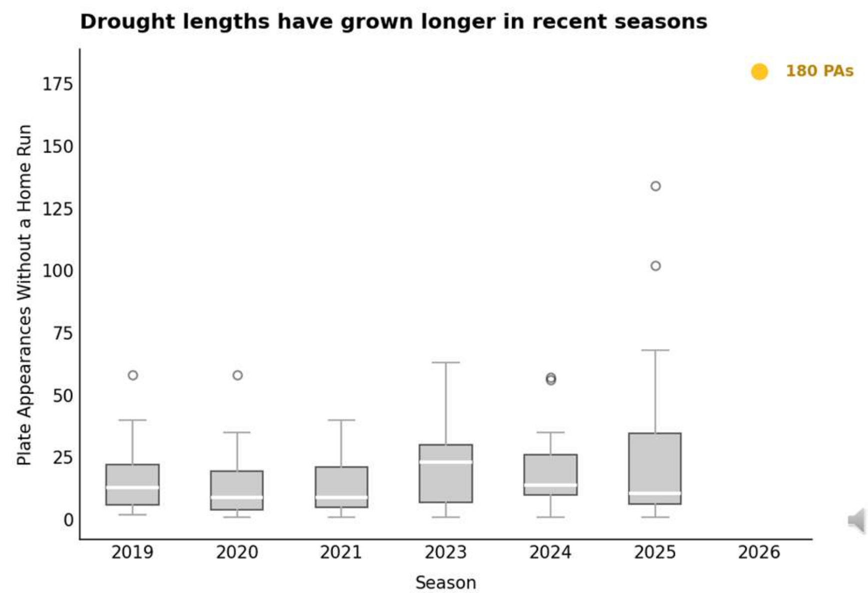
- Survival analysis models time-to-event data
- Native handling of right-censored observations; droughts still in progress aren't excluded
- Two methods: Kaplan Meier estimator for exploration, Cox Proportional Hazards model for covariate effects
- Unit of time: Plate appearances, not calendar days



The Data

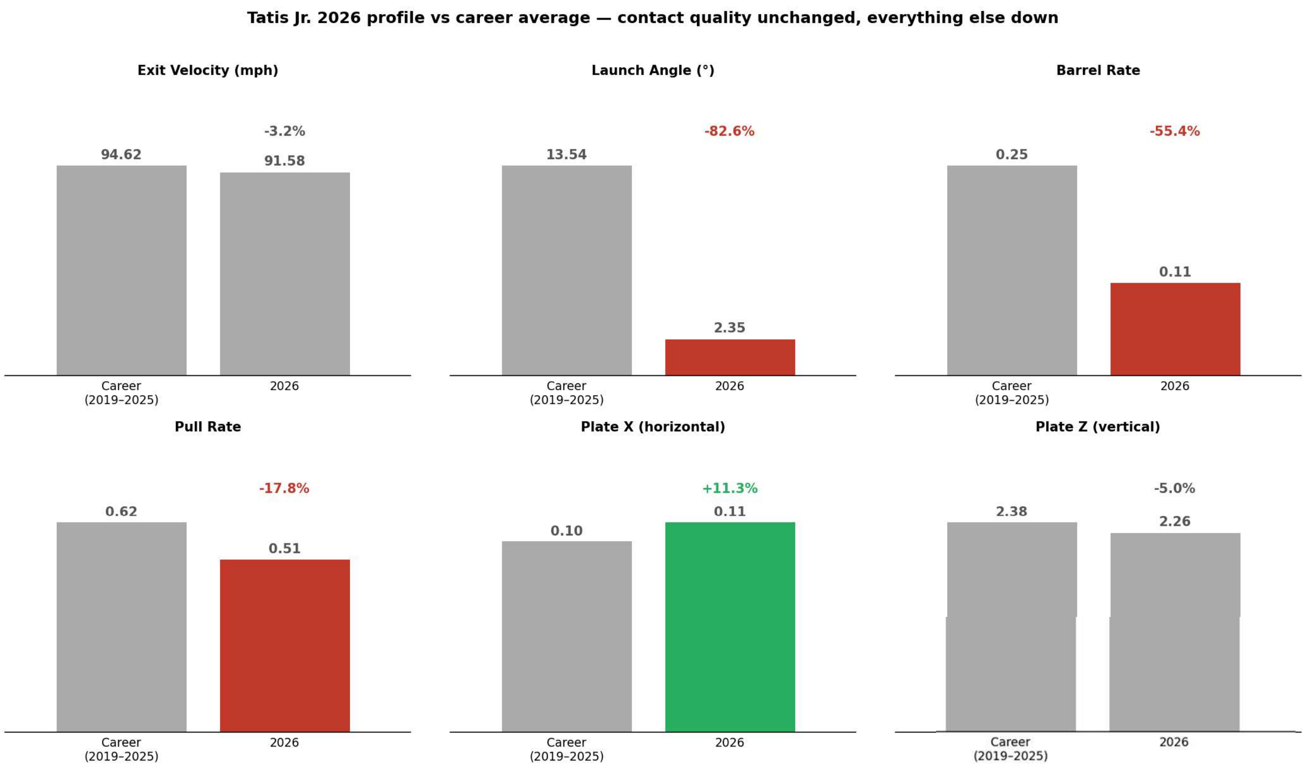
- Source: pybaseball; a Statcast API via Baseball Savant
- 3,122 plate appearances across 6 seasons (2019 – 2026)
- Filtered to regular season, one row per plate appearance
- 159 drought episodes constructed: 152 resolved, 7 censored
- Key covariates: exit velocity, launch angle, barrel rate, pull rate





What Changed in 2026

- Exit velocity: 3.2%, essentially unchanged
- Launch angle: -82.6%, from 13.5° career average to 2.35° in 2026
- Barrel rate: -55.4%, from 0.251 career average to 0.112
- Pull rate: -17.8%, career low of 0.509
- The power is there. The angle and direction are not.



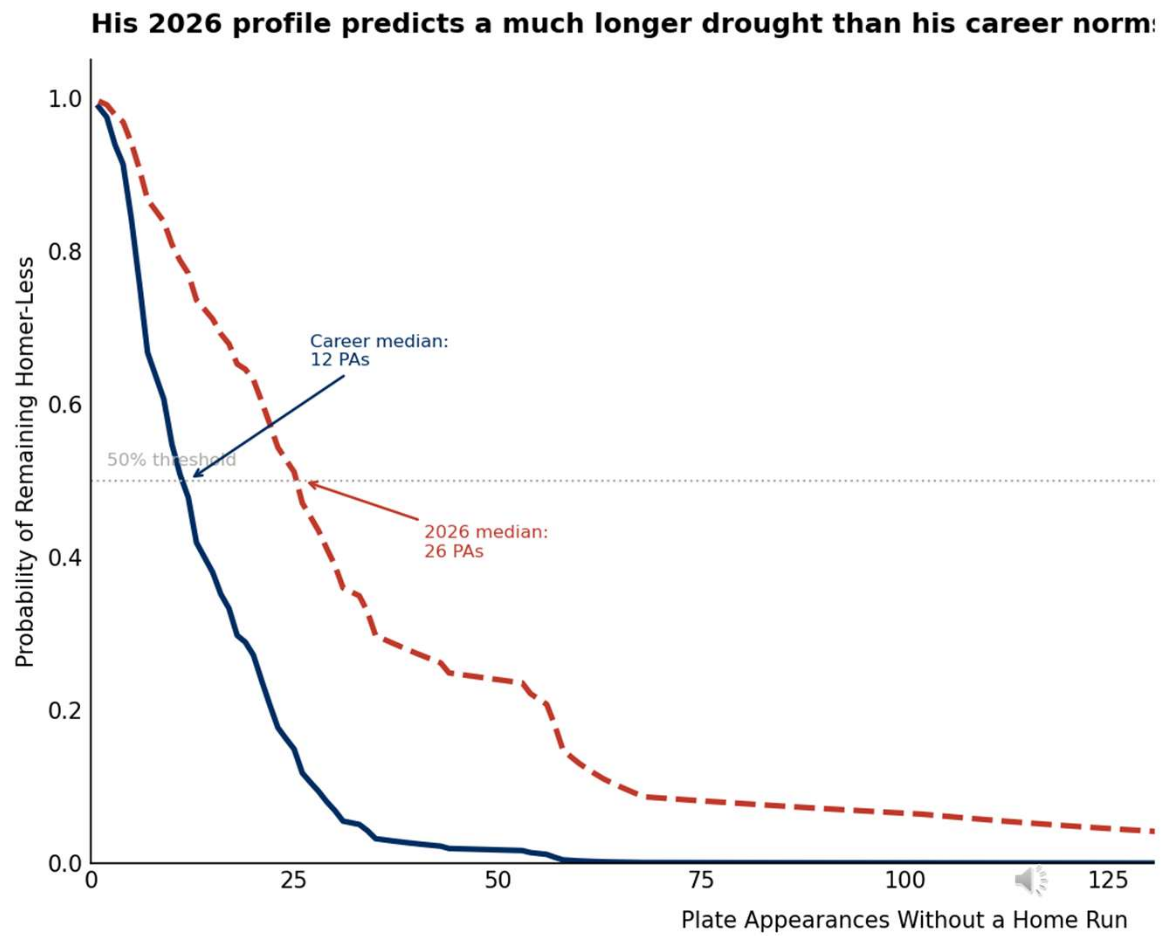
Cox Model Results

- Concordance index: 0.773
  - Model correctly ranks 77.3% of episode durations
- Barrel rate: hazard ratio = 47.17 ($p < 0.0005$
  - Dominant predictor
- Launch angle: hazard ratio = 1.031 per degree ($p = 0.003$)
  - Significant
- Proportional hazards assumption confirmed via Schoenfeld residuals
- Barreled balls convert to home runs at 50% vs 0.5% for non-barreled contact

Covariate	Coefficient	Hazard Ratio	Significant?
barrel_rate	3.854	47.17	Yes
mean_launch_angle	0.031	1.031	Yes
mean_exit_velo	0.030	1.031	No
mean_plate_z	0.351	1.420	No
pull_rate	0.246	1.278	No
mean_plate_x	-0.060	0.942	No

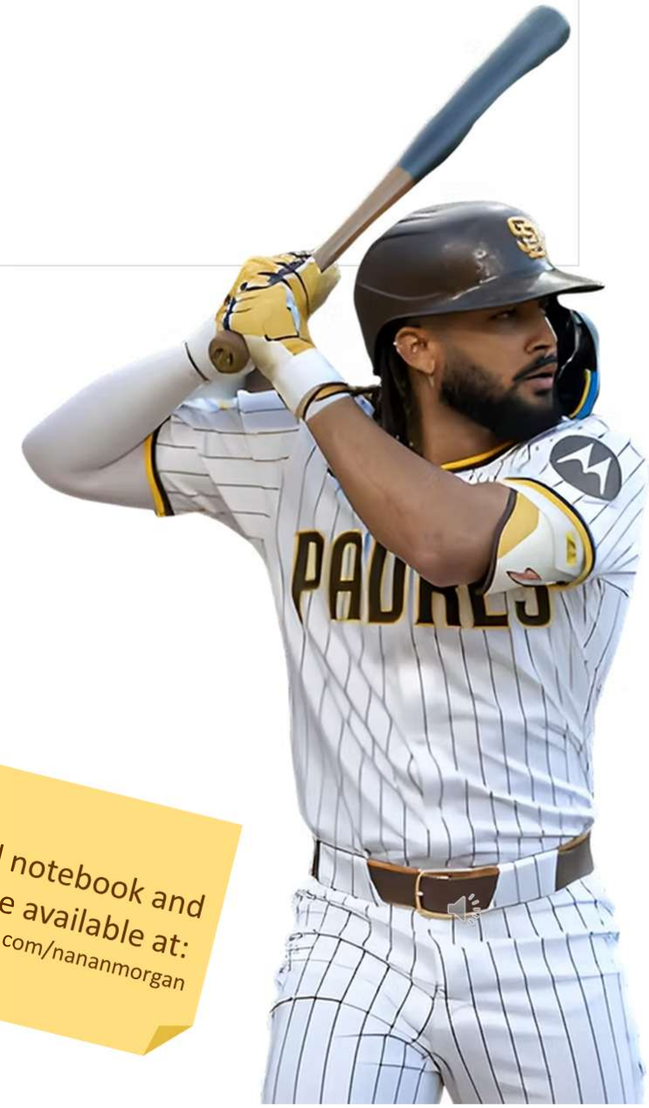
How Anomalous Is This Drought?

- KM estimate at 180 PAs: 0.85
  - Based on career drought distribution
- Cox model, career profile: 0.01%
  - Essentially impossible under historical norms
- Cox model, 2026 profile: 3.96%
  - Unusual, but profile shift explains much of it
- Current drought: 7.0 standard deviations above career mean
- Predicted median drought: 12 PAs (career profile) vs 26 PAs (2026 profile)



Takeaways and What's Next

- Survival analysis is underutilized in sports analytics
  - right-censored events are everywhere
- Watch launch angle, not exit velocity, as the leading indicator of recovery
- Framework is generalizable: any time-to-event problem with rich covariate data is a candidate
- Next steps: population model across all power hitters, interactive app, time-varying covariates



Full notebook and code available at:
github.com/nananmorgan